Problems on Infinite Series – Comparison test — with Solutions

Problems:

Test the convergence of the following series:

1. $1-\frac{1}{3}+\frac{1}{3^2}-\frac{1}{3^3}+\cdots$
2. $\frac{3}{5}+\frac{4}{5^2}+\frac{3}{5^3}+\frac{4}{5^4}+\cdots$
3. $\frac{1}{1\cdot 2}+\frac{2}{3\cdot 4}+\frac{3}{5\cdot 6}+\cdots$
4. $\frac{1}{1\cdot 3}+\frac{2}{3\cdot 5}+\frac{3}{5\cdot 7}+\cdots$
5. $\frac{2}{1^p}+\frac{3}{2^p}+\frac{4}{3^p}+\frac{5}{4^p}+\cdots$
6. $\frac{1}{1^2}+\frac{1+2}{1^2+2^2}+\frac{1+2+3}{1^2+2^2+3^2}+\cdots$
7. $\frac{1}{1\cdot 3\cdot 5}+\frac{2}{3\cdot 5\cdot 7}+\frac{3}{5\cdot 7\cdot 9}+\cdots$
8. $\frac{3}{1}+\frac{4}{8}+\frac{5}{27}+\frac{6}{64}+\cdots$
9. $\sum_{n=1}^{\infty}\frac{\sqrt{n}}{n^2+1}$
10. $\sum_{n=0}^{\infty}\frac{2n^3+5}{4n^5+1}$
11. $\sum_{n=1}^{\infty}\frac{(n+1)(n+2)}{n^2\sqrt{n}}$
12. $\sum_{n=1}^{\infty}\left(\sqrt{n^2+1}-n\right)$
13. $\sum_{n=1}^{\infty}\left(\sqrt[3]{n^3+1}-n\right)$
14. $\sum_{n=1}^{\infty}\left(\sqrt{n^4+1}-\sqrt{n^4-1}\right)$
15. $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}\sin\left(\frac{1}{n}\right)$
16. $\sum_{n=2}^{\infty}\frac{1}{\sqrt{n}\log n}$
17. $\sum_{n=1}^{\infty}\frac{\sqrt{n+1}-1}{(n+2)^3-1}$

Solutions

Problem 1

Problem Statement:
Test the convergence of the infinite series:

$$1 - \frac{1}{3} + \frac{1}{3^2} - \frac{1}{3^3} + \frac{1}{3^4} - \cdots = \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n$$

Step-by-Step Solution:
1. **Identify the Series Type:**
The series is an infinite geometric series with first term $a = 1$ and common ratio $r = -\frac{1}{3}$. 2. **Apply the Geometric Series Test:**
An infinite geometric series $\sum_{n=0}^{\infty} a\, r^n$ converges if and only if $|r| < 1$.
Here, $|r| = \left|-\frac{1}{3}\right| = \frac{1}{3} < 1$. 3. **Calculate the Sum:**
Since $|r| < 1$, the series converges absolutely to:

$$S = \frac{a}{1 - r} = \frac{1}{1 - \left(-\frac{1}{3}\right)} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

Conclusion:
The series **CONVERGES** absolutely (Sum = $\frac{3}{4}$).

Problem 2

Problem Statement:
Test the convergence of the infinite series:

$$\frac{3}{5} + \frac{4}{5^2} + \frac{3}{5^3} + \frac{4}{5^4} + \cdots$$

Step-by-Step Solution:
1. **Decompose the Series:**
Decompose the series into the sum of two separate geometric sub-series (odd-indexed and even-indexed terms):

$$\left(\frac{3}{5} + \frac{3}{5^3} + \frac{3}{5^5} + \cdots\right) + \left(\frac{4}{5^2} + \frac{4}{5^4} + \frac{4}{5^6} + \cdots\right)$$

2. **Evaluate the First Sub-Series (S_1):**

$$S_1 = \frac{3}{5} + \frac{3}{5^3} + \frac{3}{5^5} + \cdots$$

First term $a_1 = \frac{3}{5}$, common ratio $r_1 = \frac{1}{5^2} = \frac{1}{25}$.

Since $|r_1| = \frac{1}{25} < 1$, S_1 converges to:

$$S_1 = \frac{\frac{3}{5}}{1 - \frac{1}{25}} = \frac{\frac{3}{5}}{\frac{24}{25}} = \frac{3}{5} \times \frac{25}{24} = \frac{15}{24} = \frac{5}{8}$$

3. **Evaluate the Second Sub-Series (S_2):**

$$S_2 = \frac{4}{5^2} + \frac{4}{5^4} + \frac{4}{5^6} + \cdots$$

First term $a_2 = \frac{4}{25}$, common ratio $r_2 = \frac{1}{25}$.

Since $|r_2| = \frac{1}{25} < 1$, S_2 converges to:

$$S_2 = \frac{\frac{4}{25}}{1 - \frac{1}{25}} = \frac{\frac{4}{25}}{\frac{24}{25}} = \frac{4}{24} = \frac{1}{6}$$

4. **Total Sum:**

$$S = S_1 + S_2 = \frac{5}{8} + \frac{1}{6} = \frac{15 + 4}{24} = \frac{19}{24}$$

Conclusion:

The series **CONVERGES** (Sum = $\frac{19}{24}$).

Problem 3

Problem Statement:

Test the convergence of the infinite series:

$$\frac{1}{1 \cdot 2} + \frac{2}{3 \cdot 4} + \frac{3}{5 \cdot 6} + \cdots = \sum_{n=1}^{\infty} \frac{n}{(2n-1)(2n)}$$

Step-by-Step Solution:

1. **Find the General Term a_n:**

$$a_n = \frac{n}{(2n-1)(2n)} = \frac{n}{4n^2 - 2n}$$

2. **Asymptotic Behavior:**

For large n, $a_n \approx \frac{n}{4n^2} = \frac{1}{4n}$. 3. **Apply the Limit Comparison Test (LCT):**

Compare a_n with $b_n = \frac{1}{n}$ (the harmonic series):

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{n}{4n^2 - 2n}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{4n^2 - 2n} = \frac{1}{4}$$

4. **Conclusion from Test:**

Since $L = \frac{1}{4}$ is a finite positive number ($0 < L < \infty$), $\sum a_n$ and $\sum b_n$ behave identically.

The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is a p-series with $p = 1$, which **diverges**.

Conclusion:

The series **DIVERGES**.

Problem 4

Problem Statement:

Test the convergence of the infinite series:

$$\frac{1}{1 \cdot 3} + \frac{2}{3 \cdot 5} + \frac{3}{5 \cdot 7} + \cdots = \sum_{n=1}^{\infty} \frac{n}{(2n-1)(2n+1)}$$

Step-by-Step Solution:

1. **Find the General Term a_n:**

$$a_n = \frac{n}{(2n-1)(2n+1)} = \frac{n}{4n^2 - 1}$$

2. **Apply the Limit Comparison Test (LCT):**

Compare a_n with $b_n = \frac{1}{n}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{n}{4n^2-1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{4n^2-1} = \frac{1}{4}$$

3. **Conclusion from Test:**
Since $L = \frac{1}{4} > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges ($p = 1$), by the Limit Comparison Test, $\sum a_n$ also diverges.

Conclusion:
The series **DIVERGES**.

Problem 5

Problem Statement:
Test the convergence of the infinite series depending on p:

$$\frac{2}{1^p} + \frac{3}{2^p} + \frac{4}{3^p} + \frac{5}{4^p} + \cdots = \sum_{n=1}^{\infty} \frac{n+1}{n^p}$$

Step-by-Step Solution:
1. **Identify the General Term a_n:**

$$a_n = \frac{n+1}{n^p}$$

2. **Asymptotic Behavior:**
As $n \to \infty$, $a_n = \frac{n+1}{n^p} \approx \frac{n}{n^p} = \frac{1}{n^{p-1}}$. 3. **Apply the Limit Comparison Test (LCT):**
Compare a_n with $b_n = \frac{1}{n^{p-1}}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{n+1}{n^p}}{\frac{1}{n^{p-1}}} = \lim_{n\to\infty} \frac{n+1}{n} = 1$$

4. **Apply p-Series Criteria:**
The comparison series $\sum_{n=1}^{\infty} \frac{1}{n^{p-1}}$ is a standard p-series with exponent $P = p - 1$. - It converges if $P > 1 \implies p - 1 > 1 \implies p > 2$. - It diverges if $P \leq 1 \implies p - 1 \leq 1 \implies p \leq 2$.

Conclusion:
The series **CONVERGES for** $p > 2$ and **DIVERGES for** $p \leq 2$.

Problem 6

Problem Statement:
Test the convergence of the infinite series:

$$\frac{1}{1^2}+\frac{1+2}{1^2+2^2}+\frac{1+2+3}{1^2+2^2+3^2}+\cdots=\sum_{n=1}^{\infty}\frac{\sum_{k=1}^{n}k}{\sum_{k=1}^{n}k^2}$$

Step-by-Step Solution:
1. **Simplify the Numerator and Denominator:**
- Numerator (Sum of first n integers):

$$\sum_{k=1}^{n}k=\frac{n(n+1)}{2}$$

- Denominator (Sum of squares of first n integers):

$$\sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}$$

2. **Simplify the General Term a_n:**

$$a_n=\frac{\frac{n(n+1)}{2}}{\frac{n(n+1)(2n+1)}{6}}=\frac{n(n+1)}{2}\times\frac{6}{n(n+1)(2n+1)}=\frac{3}{2n+1}$$

3. **Apply the Limit Comparison Test (LCT):**
Compare a_n with $b_n=\frac{1}{n}$:

$$L=\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{\frac{3}{2n+1}}{\frac{1}{n}}=\lim_{n\to\infty}\frac{3n}{2n+1}=\frac{3}{2}$$

4. **Conclusion from Test:**
Since $L=\frac{3}{2}>0$ and $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges ($p=1$), $\sum a_n$ diverges.

Conclusion:
The series **DIVERGES**.

Problem 7

Problem Statement:
Test the convergence of the infinite series:

$$\frac{1}{1\cdot 3\cdot 5}+\frac{2}{3\cdot 5\cdot 7}+\frac{3}{5\cdot 7\cdot 9}+\cdots=\sum_{n=1}^{\infty}\frac{n}{(2n-1)(2n+1)(2n+3)}$$

Step-by-Step Solution:
1. **Find the General Term a_n:**

$$a_n=\frac{n}{(2n-1)(2n+1)(2n+3)}$$

2. **Asymptotic Analysis:**
Denominator degree is 3, numerator degree is 1.
For large n, $(2n-1)(2n+1)(2n+3)\approx 8n^3$, so $a_n\approx\frac{n}{8n^3}=\frac{1}{8n^2}$. 3. **Apply the Limit Comparison Test (LCT):**
Compare a_n with $b_n=\frac{1}{n^2}$:

$$L=\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{\frac{n}{(2n-1)(2n+1)(2n+3)}}{\frac{1}{n^2}}=\lim_{n\to\infty}\frac{n^3}{(2n-1)(2n+1)(2n+3)}=\frac{1}{8}$$

4. **Conclusion from Test:**
Since $L=\frac{1}{8}>0$ and $\sum_{n=1}^{\infty}\frac{1}{n^2}$ is a convergent p-series ($p=2>1$), $\sum a_n$ converges.

Conclusion:
The series **CONVERGES**.

Problem 8

Problem Statement:
Test the convergence of the infinite series:

$$\frac{3}{1}+\frac{4}{8}+\frac{5}{27}+\frac{6}{64}+\cdots=\sum_{n=1}^{\infty}\frac{n+2}{n^3}$$

Step-by-Step Solution:

1. **Express the General Term a_n:**

Observe that denominators are $1^3, 2^3, 3^3, 4^3, \ldots$ and numerators are 3,4,5,6, …

$$a_n = \frac{n+2}{n^3} = \frac{1}{n^2} + \frac{2}{n^3}$$

2. **Apply Series Linearity / Comparison Test:**

The series can be split into two convergent p-series:

$$\sum_{n=1}^{\infty} \frac{n+2}{n^3} = \sum_{n=1}^{\infty} \frac{1}{n^2} + 2\sum_{n=1}^{\infty} \frac{1}{n^3}$$

- $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2 > 1$ (convergent). - $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is a p-series with $p = 3 > 1$ (convergent). 3. **Conclusion:**

The sum of two convergent series is convergent.

Conclusion:

The series **CONVERGES**.

Problem 9

Problem Statement:

Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$$

Step-by-Step Solution:

1. **Identify the General Term a_n:**

$$a_n = \frac{\sqrt{n}}{n^2+1}$$

2. **Asymptotic Analysis:**

For large n, $n^2 + 1 \approx n^2$, so $a_n \approx \frac{\sqrt{n}}{n^2} = \frac{n^{1/2}}{n^2} = \frac{1}{n^{3/2}}$. 3. **Apply the Limit Comparison Test (LCT):**

Compare with $b_n = \frac{1}{n^{3/2}}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{\sqrt{n}}{n^2+1}}{\frac{1}{n^{3/2}}} = \lim_{n\to\infty} \frac{\sqrt{n}\cdot n^{3/2}}{n^2+1} = \lim_{n\to\infty} \frac{n^2}{n^2+1} = 1$$

4. **Conclusion from Test:**
Since $L = 1 > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is a p-series with $p = \frac{3}{2} > 1$, it converges. Thus $\sum a_n$ converges.

Conclusion:
The series **CONVERGES**.

Problem 10

Problem Statement:
Test the convergence of the infinite series:

$$\sum_{n=0}^{\infty} \frac{2n^3+5}{4n^5+1}$$

Step-by-Step Solution:
1. **Identify the General Term a_n:**

$$a_n = \frac{2n^3+5}{4n^5+1}$$

2. **Asymptotic Behavior:**
For large n, $a_n \approx \frac{2n^3}{4n^5} = \frac{1}{2n^2}$. 3. **Apply the Limit Comparison Test (LCT):**
Compare a_n with $b_n = \frac{1}{n^2}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{2n^3+5}{4n^5+1}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{2n^5+5n^2}{4n^5+1} = \frac{2}{4} = \frac{1}{2}$$

4. **Conclusion from Test:**
Since $L = \frac{1}{2} > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2 > 1$, it converges.

Conclusion:
The series **CONVERGES**.

Problem 11

Problem Statement:
Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \frac{(n+1)(n+2)}{n^2\sqrt{n}}$$

Step-by-Step Solution:
1. **Expand and Simplify the General Term a_n:**

$$a_n = \frac{n^2+3n+2}{n^{5/2}}$$

2. **Asymptotic Analysis:**
For large n, $a_n \approx \frac{n^2}{n^{5/2}} = \frac{1}{n^{1/2}} = \frac{1}{\sqrt{n}}$. 3. **Apply the Limit Comparison Test (LCT):**
Compare a_n with $b_n = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{n^2+3n+2}{n^{5/2}}}{\frac{1}{n^{1/2}}} = \lim_{n\to\infty} \frac{n^2+3n+2}{n^2} = 1$$

4. **Conclusion from Test:**
Since $L = 1 > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ is a p-series with $p = \frac{1}{2} \leq 1$, it diverges.

Conclusion:
The series **DIVERGES**.

Problem 12

Problem Statement:
Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \left(\sqrt{n^2+1} - n\right)$$

Step-by-Step Solution:
1. **Rationalize the Expression:**
Multiply and divide by the conjugate $\left(\sqrt{n^2+1} + n\right)$:

$$a_n = \left(\sqrt{n^2+1} - n\right) \cdot \frac{\sqrt{n^2+1}+n}{\sqrt{n^2+1}+n} = \frac{(n^2+1)-n^2}{\sqrt{n^2+1}+n} = \frac{1}{\sqrt{n^2+1}+n}$$

2. **Apply the Limit Comparison Test (LCT):**

Compare a_n with $b_n = \frac{1}{n}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{\sqrt{n^2+1}+n}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{\sqrt{n^2+1}+n} = \lim_{n\to\infty} \frac{1}{\sqrt{1+\frac{1}{n^2}}+1} = \frac{1}{2}$$

3. **Conclusion from Test:**

Since $L = \frac{1}{2} > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges ($p = 1$), the series diverges.

Conclusion:

The series **DIVERGES**.

Problem 13

Problem Statement:

Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \left(\sqrt[3]{n^3+1} - n\right)$$

Step-by-Step Solution:

1. **Use Algebraic Identity for Rationalization:**

Use $A - B = \frac{A^3 - B^3}{A^2 + AB + B^2}$ with $A = \sqrt[3]{n^3+1}$ and $B = n$:

$$a_n = \frac{(n^3+1) - n^3}{(n^3+1)^{2/3} + n(n^3+1)^{1/3} + n^2} = \frac{1}{(n^3+1)^{2/3} + n(n^3+1)^{1/3} + n^2}$$

2. **Asymptotic Behavior:**

As $n \to \infty$, $(n^3+1)^{2/3} \approx n^2$ and $n(n^3+1)^{1/3} \approx n^2$, so denominator $\approx 3n^2$. 3. **Apply the Limit Comparison Test (LCT):**

Compare with $b_n = \frac{1}{n^2}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{(n^3+1)^{2/3} + n(n^3+1)^{1/3} + n^2}}{\frac{1}{n^2}} = \frac{1}{1+1+1} = \frac{1}{3}$$

4. **Conclusion from Test:**

Since $L = \frac{1}{3} > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges ($p = 2 > 1$), the series converges.

Conclusion:

The series **CONVERGES**.

Problem 14

Problem Statement:

Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \left(\sqrt{n^4+1} - \sqrt{n^4-1}\right)$$

Step-by-Step Solution:

1. **Rationalize the Expression:**

Multiply numerator and denominator by conjugate $\left(\sqrt{n^4+1} + \sqrt{n^4-1}\right)$:

$$a_n = \frac{(n^4+1) - (n^4-1)}{\sqrt{n^4+1} + \sqrt{n^4-1}} = \frac{2}{\sqrt{n^4+1} + \sqrt{n^4-1}}$$

2. **Apply the Limit Comparison Test (LCT):**

Compare a_n with $b_n = \frac{1}{n^2}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{2}{\sqrt{n^4+1} + \sqrt{n^4-1}}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{2n^2}{\sqrt{n^4+1} + \sqrt{n^4-1}} = \frac{2}{1+1} = 1$$

3. **Conclusion from Test:**

Since $L = 1 > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges ($p = 2 > 1$), the series converges.

Conclusion:

The series **CONVERGES**.

Problem 15

Problem Statement:

Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \sin\left(\frac{1}{n}\right)$$

Step-by-Step Solution:

1. **Analyze the Behavior as $n \to \infty$:**

As $n \to \infty$, $\frac{1}{n} \to 0$. Recall $\lim_{x \to 0} \frac{\sin x}{x} = 1$, which implies $\sin\left(\frac{1}{n}\right) \approx \frac{1}{n}$.

Therefore, $a_n \approx \frac{1}{\sqrt{n}} \cdot \frac{1}{n} = \frac{1}{n^{3/2}}$. 2. **Apply the Limit Comparison Test (LCT):**

Compare a_n with $b_n = \frac{1}{n^{3/2}}$:

$$L = \lim_{n \to \infty} \frac{a_n}{b_n} = \lim_{n \to \infty} \frac{\frac{1}{\sqrt{n}} \sin\left(\frac{1}{n}\right)}{\frac{1}{n^{3/2}}} = \lim_{n \to \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

3. **Conclusion from Test:**

Since $L = 1 > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is a p-series with $p = \frac{3}{2} > 1$, it converges.

Conclusion:

The series **CONVERGES**.

Problem 16

Problem Statement:

Test the convergence of the infinite series:

$$\sum_{n=2}^{\infty} \frac{1}{\sqrt{n} \log n}$$

Step-by-Step Solution:

1. **Compare Logarithmic and Polynomial Growth:**

For $n \geq 2$, $\log n < \sqrt{n}$.

Therefore:

$$\sqrt{n} \log n < \sqrt{n} \cdot \sqrt{n} = n$$

2. **Formulate Inequality for General Term:**

Taking reciprocals reverses the inequality:

$$a_n = \frac{1}{\sqrt{n} \log n} > \frac{1}{n} \quad \text{for all } n \geq 2$$

3. **Apply the Direct Comparison Test:**

The series $\sum_{n=2}^{\infty} \frac{1}{n}$ is the harmonic series (minus its first term), which **diverges** ($p = 1$).

Since $a_n > \frac{1}{n} > 0$ for all $n \geq 2$, the series $\sum a_n$ is bounded below by a divergent series.

Conclusion:

The series **DIVERGES**.

Problem 17

Problem Statement:

Test the convergence of the infinite series:

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-1}{(n+2)^3-1}$$

Step-by-Step Solution:

1. **Analyze Dominant Terms for Large n:**

- Numerator: $\sqrt{n+1}-1 \approx \sqrt{n} = n^{1/2}$.
- Denominator: $(n+2)^3 - 1 \approx n^3$.
- General term ratio: $a_n \approx \frac{n^{1/2}}{n^3} = \frac{1}{n^{5/2}}$. 2. **Apply the Limit Comparison Test (LCT):**

Compare a_n with $b_n = \frac{1}{n^{5/2}}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{\sqrt{n+1}-1}{(n+2)^3-1}}{\frac{1}{n^{5/2}}} = \lim_{n\to\infty} \frac{\left(\sqrt{n+1}-1\right)n^{5/2}}{(n+2)^3-1} = \lim_{n\to\infty} \frac{n^3+\cdots}{n^3+\cdots} = 1$$

3. **Conclusion from Test:**

Since $L = 1 > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n^{5/2}}$ is a p-series with $p = \frac{5}{2} > 1$, it converges.

Conclusion:

The series **CONVERGES**.